# Practice workbook · Quadratic inequalities

Rearranging an inequality into the form ax^2 + bx + c < 0 (or > 0), finding the critical values and choosing the correct region.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-02

# 1. Question 1 · part b · 4 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part b only.

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**1** Find the set of values of $x$ for which

(a) $2(3x - 1) < 4 - 3x$

**(2)**

(b) $3x^2 - 8x - 3 < 0$

**(4)**

(c) **both** $2(3x - 1) < 4 - 3x$ **and** $3x^2 - 8x - 3 < 0$

**(1)**

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**(Total for Question 1 is 7 marks)**

## 1. Question 1 · part b · 4 marks · Working and answer

Working and answer

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# 2. Question 2 · part b · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

**2** $f(x) = 2x^2 - 12x + 5$

Given that $f(x)$ can be written in the form $f(x) = a(x + b)^2 + c$ where $a$, $b$ and $c$ are integers,

(a) find the value of $a$, the value of $b$ and the value of $c$

**(3)**

(b) Hence find the set of values of $x$ for which $f(x) - 37 > 0$

**(4)**

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## 2. Question 2 · part b · 4 marks

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Further practice · Attempt part b only.

**Question 2 continued**

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**(Total for Question 2 is 7 marks)**

## 2. Question 2 · part b · 4 marks · Working and answer

Working and answer

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## 3. Question 1 · part b · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

**1** Find the set of values for $x$ for which

(a) $5x - 10 > 4x - 7$

**(1)**

(b) $2x^2 - 11x + 5 < 0$

**(3)**

(c) **both** $5x - 10 > 4x - 7$ **and** $2x^2 - 11x + 5 < 0$

**(1)**

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**(Total for Question 1 is 5 marks)**

## 3. Question 1 · part b · 3 marks · Working and answer

Working and answer

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## 4. Question 2 · part b · 4 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

**2** Find the set of values of $x$ for which

(a) $3x - 8 < 5x + 3$

**(1)**

(b) $4x^2 - 7x + 1 > 6 - 2x^2$

**(4)**

(c) **both** $3x - 8 < 5x + 3$ **and** $4x^2 - 7x + 1 > 6 - 2x^2$

**(1)**

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## 4. Question 2 · part b · 4 marks · Working and answer

Working and answer

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## 5. Question 4 · part b · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

**4**

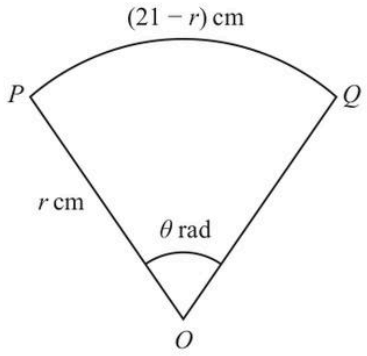



Diagram **NOT** accurately drawn

**Figure 2**

Figure 2 shows the sector $OPQ$ of a circle with centre $O$ and radius $r$ cm.

$OP = OQ = r$ cm  arc $PQ = (21 - r)$ cm  $\angle POQ = \theta$ radians

The area of the sector is $A$ cm$^2$

(a) Show that $A = \frac{r}{2}(21 - r)$ **(3)**

The area of the sector must be greater than or equal to 27 cm$^2$

(b) Find the set of possible values of $r$ **(4)**

(c) Hence write down the set of possible values of $\theta$ **(2)**

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**8**

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## 5. Question 4 · part b · 4 marks · Working and answer

Working and answer

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## 6. Question 2 · part b · 3 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

**2** Find the set of values for $x$ for which

(a) $9-3x>11x+2$

**(1)**

(b) $10x^2+7x<12$

**(3)**

(c) **both** $9-3x>11x+2$ **and** $10x^2+7x<12$

**(1)**

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## 6. Question 2 · part b · 3 marks · Working and answer

Working and answer

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## 7. Question 1 · part b · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

**1** Find the set of values for $x$ for which

(a) $8x - 7 < 5x + 5$

**(2)**

(b) $2x^2 - 5x - 3 > 0$

**(3)**

(c) **both** $8x - 7 < 5x + 5$ **and** $2x^2 - 5x - 3 > 0$

**(1)**

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**(Total for Question 1 is 6 marks)**

## 7. Question 1 · part b · 3 marks · Working and answer

Working and answer

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## 8. Question 2 · part b · 3 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

**2** Find the set of values of $x$ for which

(a) $2(x+1) < 5x - 2$

**(2)**

(b) $3x^2 - x \leqslant 10$

**(3)**

(c) **both** $2(x+1) < 5x - 2$ **and** $3x^2 - x \leqslant 10$

**(1)**

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## 8. Question 2 · part b · 3 marks · Working and answer

Working and answer

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## 9. Question 2 · part b · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

**2** Find the set of values of $x$ for which

(a) $3 + 2x \leqslant x + 2$

**(1)**

(b) $8x^2 + 10x < 3$

**(4)**

(c) **both** $3 + 2x \leqslant x + 2$ **and** $8x^2 + 10x < 3$

**(1)**

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## 9. Question 2 · part b · 4 marks · Working and answer

Working and answer

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## 10. Question 2 · part b · 2 marks

2025 NOV · Paper 02

Further practice · Attempt part b only.

**2** (a) Use algebra to find the $x$ coordinates of the points where the curve with equation $y = 3x^2 + 9x - 17$ intersects the line with equation $y = 3 - 2x$

**(3)**

(b) Hence, or otherwise, find the set of values of $x$ for which $3 - 2x \leqslant 3x^2 + 9x - 17$

**(2)**

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## 10. Question 2 · part b · 2 marks

2025 NOV · Paper 02

Further practice · Attempt part b only.

**Question 2 continued**

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**(Total for Question 2 is 5 marks)**

## 10. Question 2 · part b · 2 marks · Working and answer

Working and answer

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